Paranthesis = ( )
Curly braces = { }

8. CONDITIONAL STATEMENTS

→ 6 am → alarm

→ dant → Room saf

 → >33 → Pass

- If
- If - else
- else - if
- Ternary Operator
- Switch

If - else Statement :-

Syntax :-

```
if (condition) {---
    -----
}
else {
    ----
}
```

Note : We can use so many if statement as we want.

Ex :- Age : 18 = Vote, drive, college

```
int age = 22;
if (age >=18) {
        Sopln ("Vote, drive")
} else {
        Sopln ("Not Adult")
}
```

- Print the Largest of 2 numbers :-
  A = 1 , B = 3

code :-
```
public class Test {

    public static void main (String [] a) {
        int A = 5;
        int B = 7;

    if (A >= B) {
        Sopln ("A is largest of 2");
    }
    else {
        Sopln ("B is largest of 2");
    }
  }
}
```

= Output :- B is largest of 2

- Print if number is odd or event : divisible by 2 or multiple of 2

code :-
```
import java.util.*;

public class JavaBasics {

    public static void main (String [] a) {
```

```
Scanner sc = new Scanner (System.in);
int number = sc.nextInt();

if (number % 2 == 0) {
        Sop ln ("Even");
} else{
        Sopln("odd");
}
}
}
```

= Output:- 4
even

☐ Else Else - if :-

```
if (condition 1) {
    - - -
}
else-if (condition 2) {
    - - - - -
}
else{
    - - - - -
}
```

If true ↓ ✗

If false ↓ ✓

Income tax Calculator :-

income < 5L :- 0 % tax

income b/w 5-10 L :- 20 % tax

income > 10L :- 30 % tax

Code :-

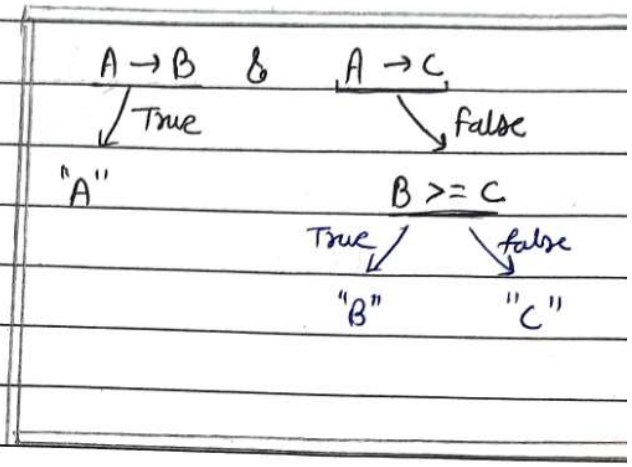
```
import java.util.*;

public class Test {
        public static void main (String [] args) {
                Scanner sc = new Scanner(System.in);
                int income = sc.nextInt();
                int tax;

Note: if use (,) ← if (income < 50000) {
in b/w zeros                    tax = 0;
Java would not          }
  understand            else - if (income >= 500000 && income < 100000) {
                                tax = (int) (income * 0.2);
                        }
                        else {
                                tax = (int) (income * 0.3);
                        }
                        Sopln(" Your tax is : " + tax);
                }
        }
```

⇒ Output : 500000
Your tax is : 120000

Print the largest of 3 numbers:-

A=1, B=3, C=6

```
if( (A>=B) && (A>=C) {
        print ("A")
  }
else-if ( B>=C){
        print ("B")
  }
else print ("C")
```



Code:-

```
import java.util.*;

public class {
    public static void main (String [] a){

        int a = 1, b=3, c=6;

        if ((a>=b) && (a>=c)){
                Sopln("a is largest");
        } else-if (b>=c) {
                Sopln("b is largest");
        } else {
                Sopln("c is largest");
        }
    }
}
```

= Output = C is largest

Ternary Operators :-

नहीं तो

Variable = Condition ? Statement 1 : Statement 2 ;

True False

int larger = (5>3) ? 5 : 3;

true

String type = (5 % 2 == 0) ? "even" : "odd" ;

Code:

```
public class Test {
    public static void main (String [] a){

        int numbers = 4;

        // ternary operators:

        String type = ((number %-2) == 0) ? "even" : "odd";

        System.out.println (type);
    }
}
```

→ Output : even

- Check if a student is Pass or Fail

marks >= 33 : Pass
marks < 33 : Fail

& code:-

```
public class Test {
    public static void main( String [] a) {
        int marks = 78 ;
        String reportCard = (marks >= 33) ? "Pass" : "fail" ;
        System.out.println( report card);
    }
}
```

→ Output : Pass

- Switch Statement :- Here switch is Keyword.

1 → Samosa
2 → Burger
3 → Mango shake

```
Switch (variable) {
        case 1:
                break;
        case 2:
                break;
        case 3:
                break;
        default:
}
```

Code:

```
public class SwitchCase {
    public static void main (String [] a) {

            int number = 2;

            Switch (number) {
                case 1 : Sopln ("Samosa");
                         break;
                case 2 : Sopln ("Burger");
                         break;
                case 3 : Sopln ("mango Juice");
                         break;
                default: Sopln ("we wake up");
            }
    }
}
```

→ Output: burger

- Calculator : +, −, *, %, /

Code :

```
import java.util.*;

public class Test{
    public static void main (String [] args){

        Scanner sc = new Scanner (System.in);
        Sopln("Enter a:");
        int a = sc.nextInt();
        Sopln("Enter b:");
        int b = sc.nextInt();
        Sopln("Enter Operator");
        Char operator = sc.next().charAt(0);

        switch (operator){

            case '+' : Sopln (a+b);
                break;
            case '-' : Sopln(a-b);
                break;
            case '*' : Sopln(a*b);
                break;
            case '/' : Sopln(a/b);
                break;
            case '%' : Sopln(a%b);

            default : Sopln("wrong operator");
        }
    }
}
```

Output :-

```
Enter a:
5
Enter b:
10
Enter operator
*
50
```